

equally willing to buy or sell at this rate as long as invariant k is constant.

Consider a concrete example (Figure 6.6). For simplicity, we will ignore transaction fees (gas) in all of the examples. Assume an investor in the Uniswap USDC/DAI market has 4 DAI (Asset A) and 4 USDC (Asset B). This sets the instantaneous exchange rate at 1 DAI:1 USDC and the invariant at 16 ($= x*y$). To sell 4 DAI for USDC, the investor deposits 4 DAI to the contract and withdraws 2 USDC. Now the USDC balance is $4 - 2 = 2$ and the DAI balance is $4 + 4 = 8$. The invariant remains constant at 16. Notice that the effective exchange rate is now 2 DAI: 1 USDC. The change in the exchange rate is due to slippage because of the low level of liquidity in the market. The magnitude of the invariant determines the amount of slippage. To extend the example, assume the balance is 100 DAI and 100 USDC in the contract. Now the invariant is 10,000, but the exchange rate is the same. If the investor sells 4 DAI for USDC, now 3.85 USDC can be withdrawn to keep the invariant constant and results in much lower slippage at an effective rate of 1.04 DAI: 1 USDC.

Deep liquidity helps minimize slippage. Therefore, it is important that Uniswap incentivizes depositors to supply capital to a given market. Anyone can become a liquidity provider by supplying assets on both sides of a market at the current exchange rate.¹³ Supplying both sides increases the product of the amount of assets held in the trading pair (i.e., increases the invariant as mentioned in the formula for